GGG-B-175A
October 20, 1987
SUPERSEDING
GGG-B-175
February 28, 1957

FEDERAL SPECIFICATION

BENDER, ELECTRICAL CONDUIT, HAND (PORTABLE)

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers portable hand tools for bending standard wall (heavy) rigid steel conduit and electrical metallic tubing.

1.2 Classification.

1.2.1 Types and sizes. The benders covered by this specification shall be of the following types and sizes, as specified (see 6.2).

Type I - Standard wall (heavy) rigid steel conduit
Size 3/4 inch - For 1/2 and 3/4 inch conduit
Size 1-1/4 inch - For 1 and 1-1/4 inch conduit

Type II - Electrical metallic tubing
Size 1/2 inch - For 1/2 inch trade size
Size 3/4 inch - For 3/4 inch trade size
Size 1 inch - For 1 inch trade size

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:
PPP-P-40 - Packaging and Packing of Hand Tools

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; Los Angeles; San Francisco; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their Agencies.)

Military Standard:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or solicitation for offers, shall apply.

American National Standards Institute (ANSI):

B1.20.3 - Dryseal Pipe Threads
NFPA 70 - National Electrical Codes

(Application for copies should be addressed to The American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for convenience of identification and are not intended to preclude the purchase of benders otherwise in accordance with this specification.

3.2 Material. Material used in construction of benders shall conform in composition, and heat-treatment to the requirements specified herein.

3.3 Treating and painting. The benders shall be treated and painted in accordance with the manufacturer's standard commercial practice.

3.4 Design. The benders shall be capable of bending the specified size of conduit or tubing to the minimum radius of the curve of the inner edge permitted by the National Electrical Code (NFPA 70) as follows:

4-inch radius for 1/2 inch size
5-inch radius for 3/4 inch size
6-inch radius for 1 inch size
8-inch radius for 1-1/4 inch size

Benders shall be so designed that conduit or tubing may be inserted from the side of the tool.

3.5 Type I, standard wall (heavy) rigid steel conduit. Type I benders shall be a one-piece hand tool consisting of a curved fulcrum, an extended jaw to grip the conduit, and a stud or socket for attachment of an ordinary pipe handle, and shall be similar to figure 1.

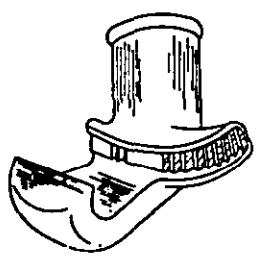


FIGURE 1. Type I bender for standard wall (heavy) rigid steel conduit.

3.5.1 Construction. The benders shall be of steel, malleable iron, ductile iron or alloy A356 aluminum, heat-treated to a T-6 condition to prevent deformation during maximum use. The handle mounting stud shall be threaded internally or externally with American Standard Taper Pipe conforming to ANSI B1.20.3. Handle mounting threads shall be 1 inch minimum for the 3/4 inch size and 1-1/2 inches for the 1-1/4 inch size bender. At the manufacturer's option, provision may be made of securing the handle by setscrews, in lieu of threading the handle mounting stud or socket.

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3.6 Type II, electrical metallic tubing. Type II benders shall be a hand tool consisting of a curved fulcrum, an extended jaw to grip the tubing, and a stud or socket for attachment of an ordinary pipe handle, and shall be similar to figure 2.

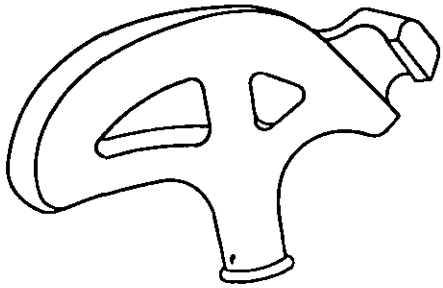


FIGURE 2. Type II bender for electrical metallic tubing.

3.6.1 Construction. The benders shall be of steel, malleable iron, ductile iron, alloy A356 aluminum, heat-treated to a T-6 condition or alloy A380 aluminum with strength to prevent deformation during maximum use. The curved fulcrum shall support the tubing to prevent crushing or reducing the inside diameter of the tubing more than 10 percent. The benders shall be capable of bending the tubing and shall not damage the surface of galvanized tubing in making bends up to 90 degrees. The handle-mounting stud shall be threaded internally or externally with American Standard Taper Pipe conforming to ANSI B1.20.3, except that for the 1 inch minimum size the handle may be secured by setscrews or any other acceptable industry method. Handle mounting threads shall be 3/4 inch for the 1/2 inch size, 1 inch for the 3/4 inch and 1 inch size benders. At the manufacturer's option, provision may be made for securing the handle mounting stud or socket. When specified, symbols or bench marks shall be cast into the benders to enable accurate 30, 45, and 90 degrees bends.

3.7 Handles. Handles, when specified (see 6.2), shall be made of galvanized steel pipe threads as stated in 3.5.1 and 3.6.1. Thread sizes and length of handles shall be as specified in table I. A galvanized steel threaded sleeve shall be furnished with each handle when bender has a male thread.

TABLE I. Thread sizes and handle length

Type	Conduit pipe size	Handle thread size	Length of pipe
	Inches	Inches	Feet
I	3/4	1	4
I	1-1/4	1-1/2	6
II	1/2	3/4	3
II	3/4	1	4
II	1	1	5

3.8 Identification markings. Each bender shall be etched or stamped in a permanent and legible manner with the manufacturer's name or identifying symbol of such known character that the source may be readily identifiable, part number and country of origin.

3.9 Workmanship. The benders shall be free from any defects that would affect appearance, serviceability or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for performance of

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the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that the supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and testing. Sampling shall be in accordance with MIL-STD-105. The lot size shall consist of all benders offered for acceptance at one time. The sample unit for end item inspection and testing shall be one bender. The sample unit for preparation for delivery inspection shall be one container fully prepared for delivery.

4.3 Inspection of the end item. The benders shall be inspected for treatment and painting, design, construction, workmanship, marking and the dimensional requirements as specified herein. The inspection level shall be I with an acceptable quality level (AQL) of 2.5 percent defective.

4.4 Testing of the end item. Each sample of type I benders shall make at least five 90-degree bends on standard wall (heavy) steel conduit of the applicable trade size. Each sample of type II benders shall make at least five 90-degree bends on electric metallic tubing of the applicable trade size. The inspection level shall be S-2 with an AQL of 4.0 percent defective.

4.5 Inspection of preparation for delivery. An examination shall be made to determine compliance with the requirements of section 5. Defects shall be scored in accordance with table II. The inspection level shall be S-2 with an AQL of 4.0 percent defective.

TABLE II. Classification of Preparation For Delivery Defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging and packing shall be A or C as specified (see 6.2).

6. NOTES

6.1 Intended use. Type I benders are used in making offset and 90-degree bends on standard wall (heavy) rigid steel conduit required for certain installations or unusual locations. Type II benders are used in making offset and 90-degree bends on electrical metallic tubing (thin wall) used currently in most installations.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

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(a) Title, number, and date of this specification.
(b) Type and size of bender required (see 1.2).
(c) Whether symbols or bench marks are required (see 3.6.1).
(d) Specify with or without handles as required (see 3.7).
(e) Level of preservation, packaging and packing required (see 5.1).

MILITARY INTERESTS

PREPARING ACTIVITY

GSA - FSS

Military Coordinating Activity

Army - GL

Custodians

Army - GL
Air Force - 99

Review activity

Air Force - 84

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